

Digital Electronics Lab Report

Submitted by : Neelesh Dorairaj

Roll number : 24EE10015

Lab code : EE29210

Date of experiment : 17-03-26

Experiment 8 : Rolling display of a student's roll number on a 7-segment display

Objectives:

1. To store multiple digits of a roll number using registers.
2. To display the digits sequentially on a 7-segment display using multiplexing.

Apparatus:

- IC-74374 Octal Register (3)
- IC-74157 Quad Multiplexer (1)

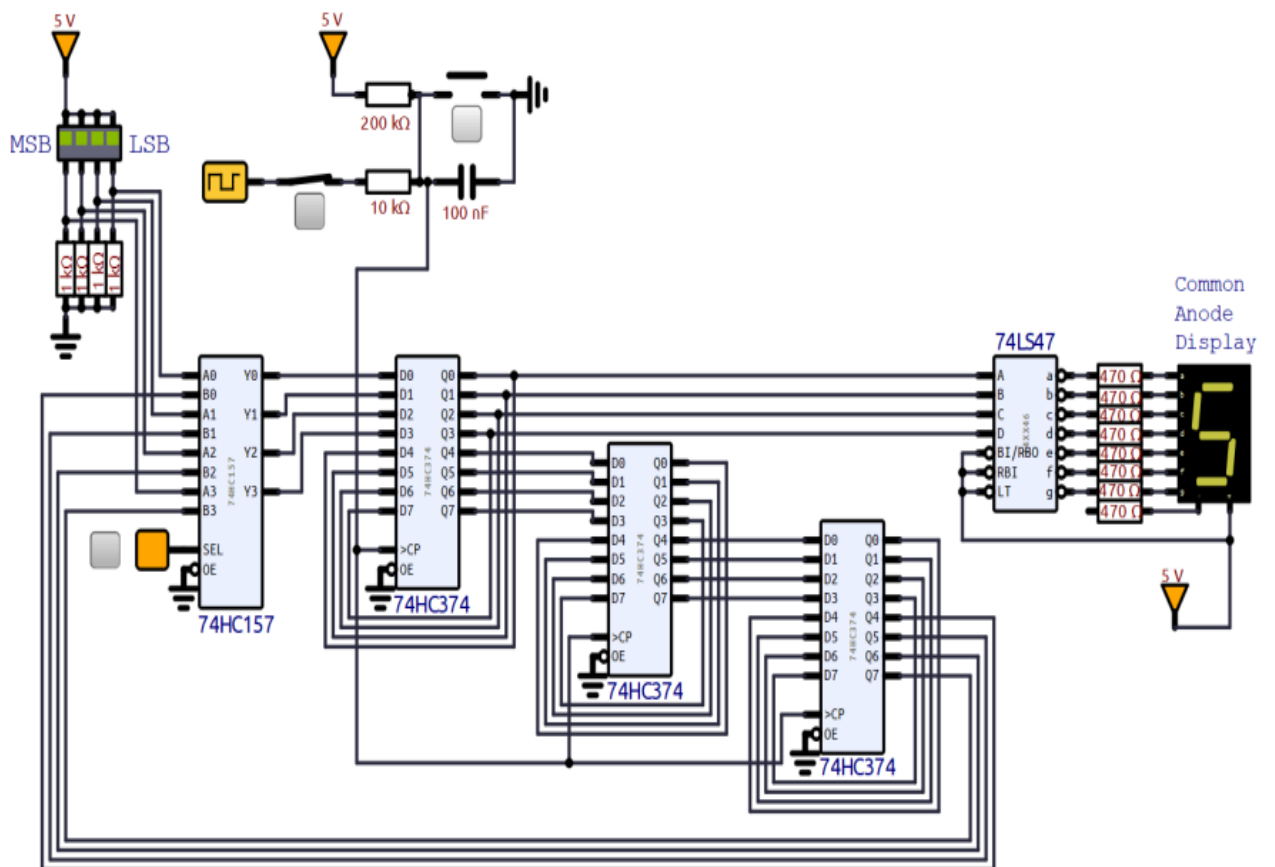
- IC-7447 BCD to 7-segment decoder (1)
- 7-segment LED display (Common anode) (1)
- Resistor 470 Ω or 390 Ω (8)
- Resistor 1 k Ω (4)
- Resistor 200 k Ω (1)
- Resistor 10 k Ω (1)
- Capacitor 100 nF (1)
- Push Button (1)
- Breadboard
- DIP Switch (2)
- Connecting wires
- DC Power Supply 5V (1)
- Function Generator (Square wave, 1 Hz) (1)

Theory:

This experiment demonstrates the rolling display of a multi-digit number using digital logic circuits. The digits of the roll number are stored in registers (IC-74374). Each register holds a 4-bit BCD value corresponding to a

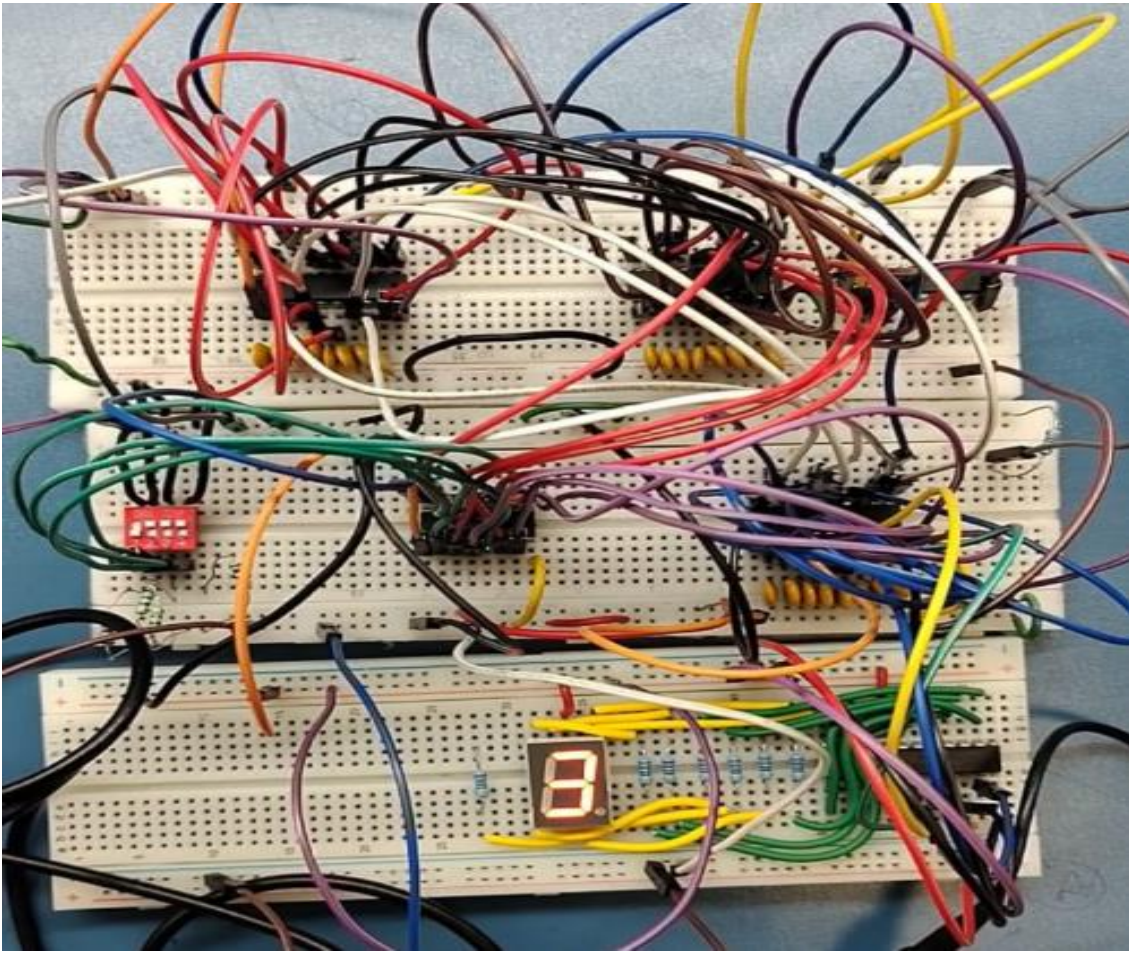
decimal digit. A multiplexer (IC-74157) selects one digit at a time, and the selected output is fed into a BCD to 7-segment decoder (IC-7447) to drive the display. A clock signal from a function generator changes the selected digit periodically, producing a rolling display due to persistence of vision.

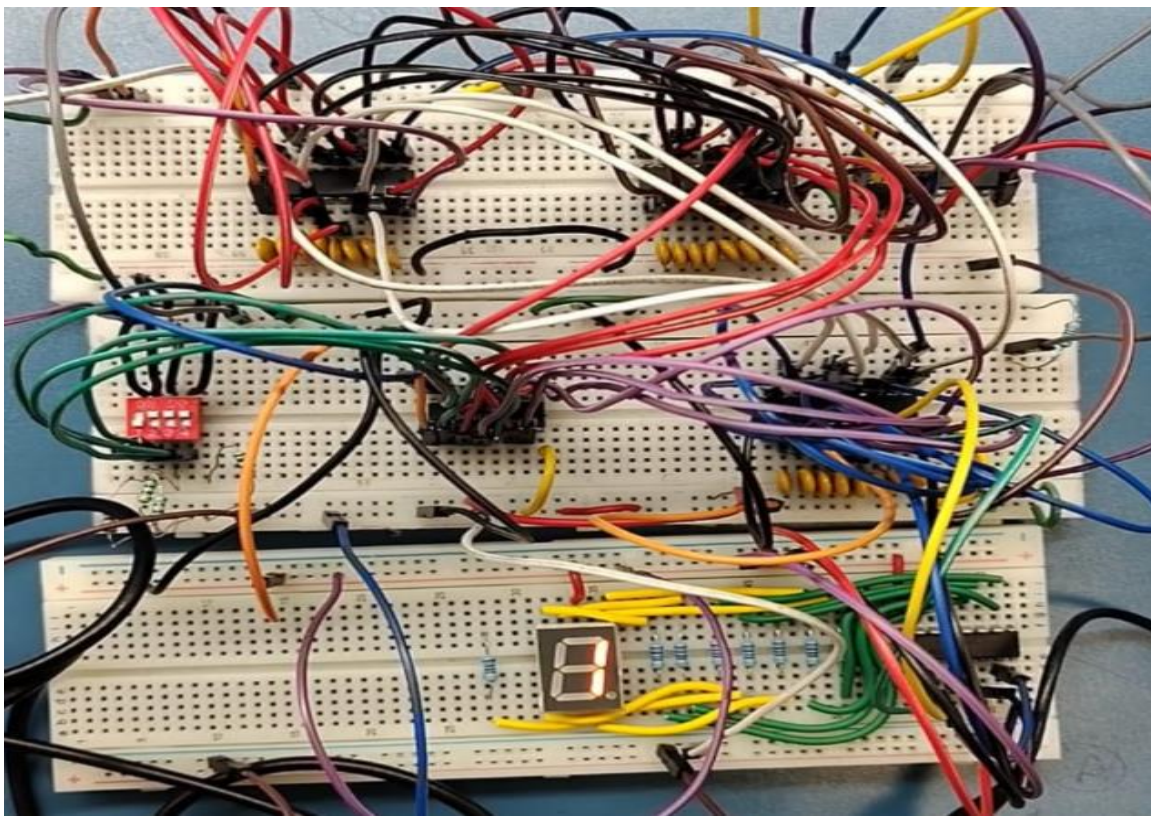
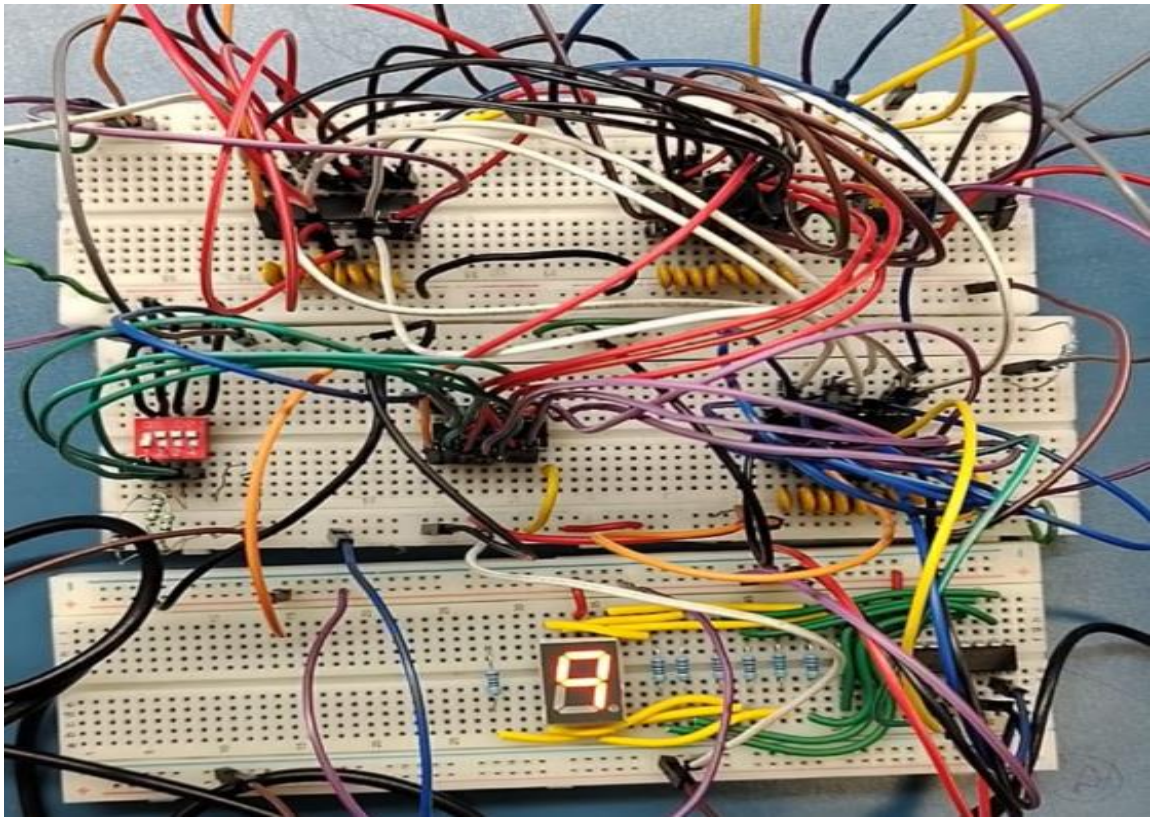
Circuit Diagram:

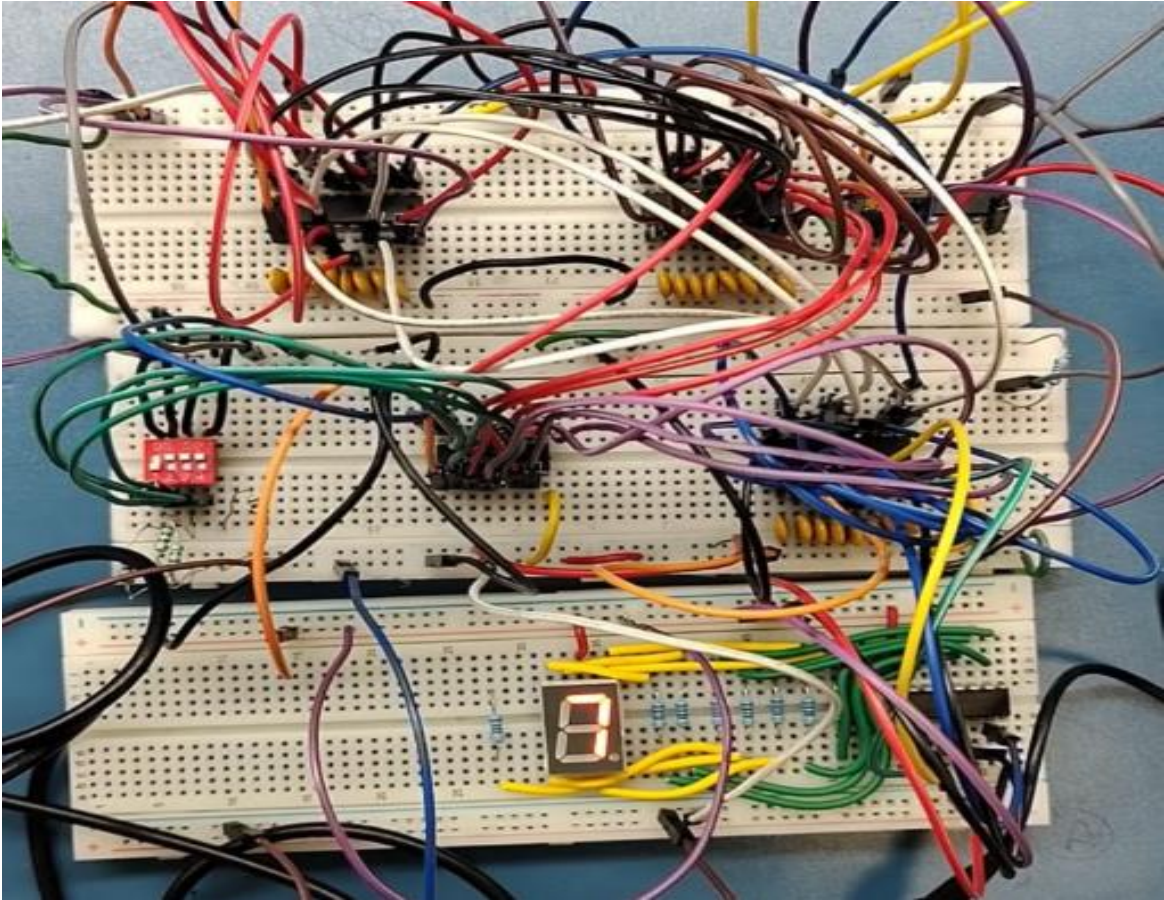


Output/Observation:

1. Each digit was stored successfully using the push button.
2. The display cycled when clock was applied.
3. Sequential display observed at ~ 1 Hz.
4. Blank state observed where required.
5. Rolling effect clearly visible.







Sequence : 3917

Conclusion:

The rolling display was successfully implemented using registers, multiplexer, and decoder. The experiment demonstrated sequential data display using clock control and provided understanding of multiplexing and display systems.